

DESIGN PROJECT REPORT

ENGR 131- Transforming Ideas to Innovation I

Section LC03

Schleman Forney Intersection Campus Safety Project



Schleman Forney Intersection as Seen From Birds Eye View (Google Earth 2022)

Team 08

Names:

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10/28/2022

I. EXECUTIVE SUMMARY

Students across campus continuously encounter the intersection bounded by the engineering mall and the engineering fountain. Students and staff (users) crossing this intersection are not just on foot; they travel by bike, skateboard, or on an assortment of electric vehicles. Purdue students, staff, and the Purdue EMS are the clients because they are the ones who will save money and time when there are fewer injuries. This poses a problem for travelers as there are no designated bike lanes or walking paths, which compels travelers to use a random direction, causing danger when numerous travelers cross simultaneously. Regarding the students' safety and the intersection's problems, the design of this intersection needs to be adjusted and fixed to create a safer area of travel for Purdue's students and staff. Our criteria that we must uphold in our project include that it should be affordable, safe, and weather resistant. Our constraints are that the product must reduce accidents. This constraint is measurable because it is possible to monitor the number of accidents both before and after the proposed bike lane is installed. Purdue University (stakeholder) needs a solution to this intersection problem. A safer intersection would help Purdue Universities' environment because fewer people would get hurt. This would also lead to fewer ambulance calls, reducing emissions of the 911 motorcade and helping the environment. Also, Purdue will be a safer place to live in, with fewer students getting injured. Our design implementation consists of the installation of barriers in a strategic manner that allows for the intersection by the engineering fountain to be safer for both students and staff. Barriers are implemented throughout areas and intersections all over the world. What makes our barriers different from some of these other barriers is how they are placed. First, we wanted our barriers to be specific to Purdue University. So our barriers will be colored in Purdue University colors to represent the student body and pride. Additionally, and as we have stated before, our barriers will be placed so they are not an interruption to the flow of traffic, just set so that they divide bikers and skateboarders from pedestrians. Many barriers worldwide are placed to disrupt and stop traffic flow, but ours are not implemented for that reason. The limitations of the solution are that the product will slowly degrade over time and will need to be replaced. The other limitation is that there is no active discouragement for bikers to stay out of the pedestrian lane. This could cause additional traffic in a smaller area, which would be even more dangerous.

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II. TEAM MEMBER ROLES

Our project team included 4 members (See Figure 1). We allocated roles and responsibilities to each team member by/based on

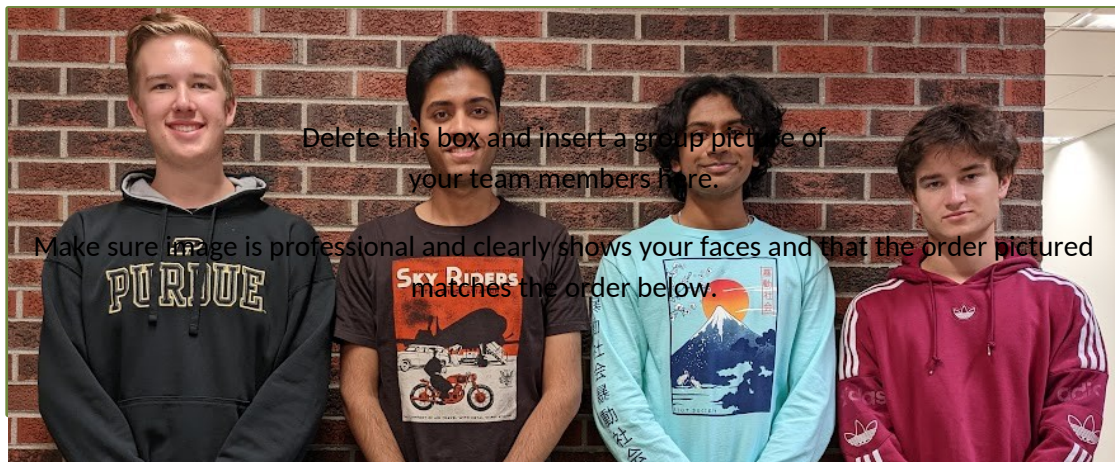
Table 1. Team roles

Project milestones	Name	Specific responsibilities related to each milestone
Problem scoping	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Problem Statement ▪ Direct Users and Stakeholders ▪ Criteria and Constraints ▪ Problem Statement
Idea generation	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Direct Users and Stakeholders Physical Model & Other Generation Ideas ▪ drawings of ideas ▪ Generating ideas by exploring previous art ▪ Flow Chart, Table 5 #3
Prototyping	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Prototype 1 & 2 and Various Writeups ▪ Weighted Decision Matrix ▪ Prototype 3 ▪ Weighted Decision Matrix
Testing	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Prototype 1 & 2 and Various Writeups ▪ Weighted Decision Matrix ▪ Prototype 3

		▪ Provided feedback on Prototype 3
WDM	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Cleaned up WDM writeup ▪ Weighted Decision Matrix ▪ Double checked WDM ▪ Weighted Decision Matrix
Preliminary presentation	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Presented in front of the groups ▪ Presented in front of the groups ▪ Presented in front of the groups
Iteration	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Wrote two pieces of feedback and reflection ▪ Wrote one piece of feedback ▪ Wrote one piece of feedback ▪ Wrote one piece of feedback
Final report	James Mareachen Soumil Iyer Dhruv Srivastava Sasha Ilijevski	▪ Wrote Executive summary, part f, CAD Drawings ▪ Wrote part E ▪ Wrote part C ▪ Wrote part B
Final presentation	James Mareachen Soumil Iyer Dhruv Srivastava	▪ Worked on 3 slides ▪ Worked on 3 slides

	Sasha Ilijevski	▪ Worked on 3 slides ▪ Worked on 3 slides
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Figure 1. Project team



III. PROBLEM SCOPING

a. Problem Statement

Students across campus continuously encounter the intersection bounded by the engineering mall and the engineering fountain. Students and staff (users) crossing this intersection are not just on foot, they are traveling by bike, skateboard, or on an assortment of electric vehicles. Purdue students, staff, and the Purdue EMS are the clients due to the fact that they are the ones who will save money and time when there are fewer injuries. This poses a problem for travelers as there are no designated bike lanes and no designated walking paths which compel travelers to just use a random path, causing danger when numerous travelers are crossing at the same time. In regards to the student's safety and the problems of the intersection, the design of this intersection needs to be adjusted and/or fixed to create a safer area of travel for Purdue's students and staff. Purdue University (stakeholder) needs a solution to this intersection problem. The implementation of a safer intersection would help the Purdue society because fewer people would get hurt. This would also lead to fewer ambulance calls, reducing emissions of the 911 motorcade, and helping the environment as well. Additionally, with fewer students getting injured, Purdue will simply be a safer society to live in as well.

b. Design Criteria and Constraints

Table 2. Design requirements

Criteria and Constraints	Metrics (ways to quantify and measure the performance of your solutions and measure success)	Metric units
Minimum Cost	Summing the total cost of the project, and attempting to decrease this cost as much as possible while maintaining the integrity of the project.	US Dollars
Reduces Accidents	Place a camera that watches the intersection, and observe the number of accidents that occur before and after the installation is put in place.	Number of accidents
User Friendly	Survey the users of the project before and after its implementation to see whether or not the design is satisfactory or not. The data can be collected on a scale from 1 through 10, to quantify the data.	The average number on the scale
Does not negatively affect the flow of traffic	The typical time spent in traffic could be recorded before and after the implementation of the solution. The solution should not impede traffic, so the second time recorded must be equal to or less than the first time.	Seconds

Trade-off Considerations

Trade-off 1:

Having a proper intersection can slow down traffic when there are too many of one type of traveler, as having a divided bike lane and walking lane decreases the effective space for both users.

Trade-off 2:

By creating a new bike lane, more money needs to be spent overall, as every few years it will need to be repainted, but there will be a higher amount of safety.

Trade-off 3:

The painting process will slow down traffic for a few days each year, therefore, to increase the speed of traffic overall, the traffic will need to slow down every year for a few weeks to be painted.

Trade-off 4:

It will be harder to cross between buildings because you will have to go through the bike lane to cross the road. Biking and walking straight will be better though, as the flow of traffic will be better.

c. Direct Users and Stakeholders

Table 3. Empathizing with users

User segment (include an image and a label)	Methods used to empathize with the user (interview user, survey user, observe user, read about user, simulate user behavior, put yourself in his/her shoes)	Lessons learned from this interaction
	- Interviews - Online surveys	The various bikers and skateboarders disliked the lack of clarity that the paths give. They would much prefer if they were able to have paths for just them and

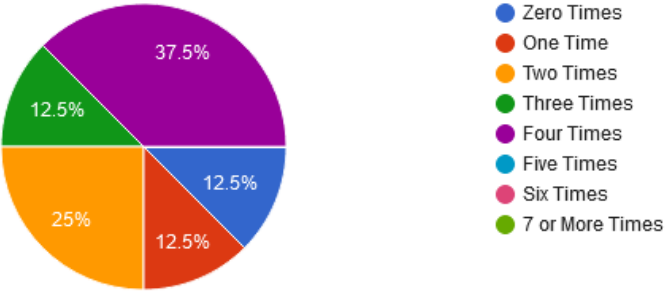
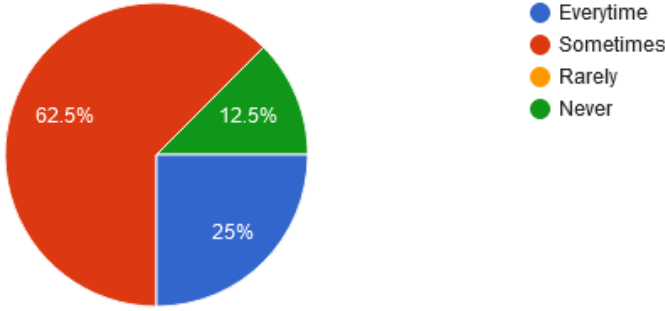
People at Purdue traveling by skateboards, bikes, segways etc. (Bangert, 2019)		clear places where they are able to ride and stay safe.
 People at Purdue traveling by foot (Purdue University, 2022)	- Interviews - Online surveys	Pedestrians at Purdue would like to see clearer pathways. Constantly avoiding bikes and swerving so that they are not ran over is quite unsafe. They would prefer if bikers would stay in their own lanes.
 Director of Treasury Operations at Purdue University (Terri Mimms, 2022)	- Interview - Survey - Read about her on Purdue website	Having to spend less money on 911 calls and medical expenses for those injured is a venture that is quite profitable for Purdue.

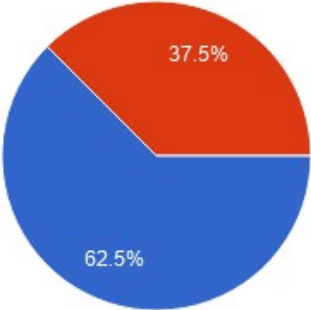
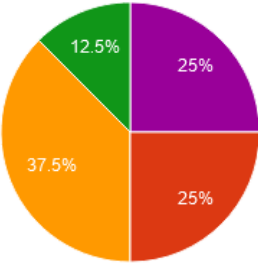
d. Background Information

The Schleman Forney intersection has always been unsafe. Set in the center of the engineering campus, there is constant traffic of bikers, skateboarders, and pedestrians, and having no clear path makers for either makes it very unsafe. Adding bike lanes or any other type of intersection will allow for safer and quicker pathways for both pedestrians and alternative vehicular options. Creating such pathways for quick safe transportation is something that has occurred throughout campus but has stopped near the engineering fountain for unknown reasons.

Table 4. Information gathering for problem scoping

Questions asked	Answers to questions
● How often do you walk	Four Times occurred the most often, as this pathway is quite popular. there was a spread though, as many people walk

through this intersection each day?	through this intersection different amounts of times. We conducted an anonymous survey of over 30 Purdue college students and calculated these percentages based on their responses.  <table border="1"> <thead> <tr> <th>Frequency</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Zero Times</td> <td>12.5%</td> </tr> <tr> <td>One Time</td> <td>12.5%</td> </tr> <tr> <td>Two Times</td> <td>25%</td> </tr> <tr> <td>Three Times</td> <td>12.5%</td> </tr> <tr> <td>Four Times</td> <td>37.5%</td> </tr> <tr> <td>Five Times</td> <td>0%</td> </tr> <tr> <td>Six Times</td> <td>0%</td> </tr> <tr> <td>7 or More Times</td> <td>0%</td> </tr> </tbody> </table>	Frequency	Percentage	Zero Times	12.5%	One Time	12.5%	Two Times	25%	Three Times	12.5%	Four Times	37.5%	Five Times	0%	Six Times	0%	7 or More Times	0%
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Four Times	37.5%																		
Five Times	0%																		
Six Times	0%																		
7 or More Times	0%																		
● How frequently do you have to stop or slow down and wait for bikers or skateboarders to pass?	Everytime and Sometimes were the most frequent responses, the traffic there is quite congested so this correlation of data makes sense, as the intersection is not ideal. We conducted an anonymous survey of over 30 Purdue college students and calculated these percentages based on their responses.  <table border="1"> <thead> <tr> <th>Frequency</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Everytime</td> <td>25%</td> </tr> <tr> <td>Sometimes</td> <td>62.5%</td> </tr> <tr> <td>Rarely</td> <td>0%</td> </tr> <tr> <td>Never</td> <td>12.5%</td> </tr> </tbody> </table>	Frequency	Percentage	Everytime	25%	Sometimes	62.5%	Rarely	0%	Never	12.5%								
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● Have you ever been hit or almost hit by a vehicle/bike when walking through this intersection? (Followup) ● How often do you find yourself in this situation?	66.7% of people said yes, 33.3% of people said no in our anonymous survey of over 30 Purdue students.																		

	 62.5% 37.5% ● Yes ● No of the people that said yes, the majority said that they saw themselves in this position somewhat frequently. We conducted an anonymous survey of over 30 Purdue college students and calculated these percentages based on their responses.  37.5% 25% 25% 12.5% ● Very Frequently ● Frequently ● Somewhat Frequently ● Infrequently ● Never
● What improvements can be made to this intersection?	add a designated bike lane for travelers. define the difference between a bike lane and walk path Moving the bike lane so that it is not directly in front of the exit of the fountain. allowing a smoother transition from the bike lane to the pedestrain lane bike lane People just need to pay attention Through our collected responses of 30 Purdue students, we were able to figure out their opinions on the intersection and how to improve it. most believed that a bike lane should have a clearer beginning and end, as it is quite segmented currently.

e. Assumptions about the Problem

We are assuming that adding additional pieces of infrastructure will improve the flow of the intersection. Another assumption is that the intersection has no existing contracts or ownership that will prevent the addition of new street paint or signs. We are assuming that the intersection does pose a real risk for injury and accidents between travelers. We are also assuming that the people we interviewed held similar opinions to the whole population of their respective background.

IV. IDEA GENERATION & THOUGHT EXPERIMENTS

a. Functional Decomposition

<i>Main function</i>	<i>Sub-functions</i>	<i>Design ideas that meet this function</i>
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Figure 2. Functional decomposition for a new Designed Fountain Intersection

b. Exploring Prior Art

Table 5. Generating ideas by exploring prior art

#	Image/Picture & Name of Prior Art	Strengths	Weaknesses	Improved alternative
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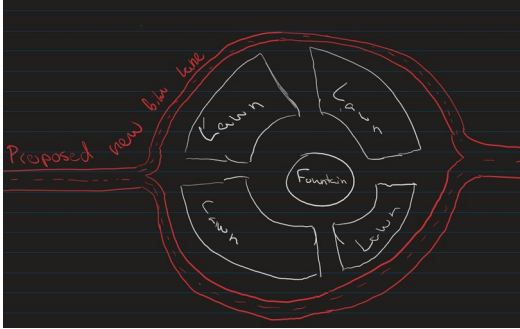
1	 <i>New pop-up bike lanes</i> (City Planning, 2022) Pop-up bike lanes: The orange-colored barricade can come up to allow bikers, skateboarders, and Segways to pass through when the traffic light is green. When the traffic light is red and pedestrians have to walk by, the barricades can go down.	● Ensures safety of both, pedestrians and bikers/skateboarders /segways	● Requires re-construction near the Engineering Fountain which can be expensive and prove to be an inconvenience for students and staff.	permanent barrier lanes. would be constructed quickly, requiring no wiring or other complex tasks to occur for proper function.
2	 <i>Small dividing lane bumps</i> (Luke, 2021) Small permanent barricades: This separates the bike lane hence avoids direct interaction between pedestrians and bikers.	● Ensures safety of both, pedestrians and bikers/skateboarders /segways ● Relatively inexpensive	● easily broken ● will wear out quickly with time ● will most likely prove to be a large hindrance when covered with snow	a tall plastic permanent barrier that shows a clear difference between the bike and person lane. prevents them from being mixed, and does not hide when there is a foot of snow on the ground.
3	Speed Bumps in bike lane: This forces cyclists/skateboarders to slow down to safer speeds if traveling too fast.	● Keeps pedestrians/skateboarders/cyclists safer in the event of a collision because of slower speeds.	● Can be potentially dangerous for skateboarders (may fall off) as it reduces	Instead of large speed bumps in the bike lane, the installation of grooves within the space of travel would be more effective and safe.

	 <i>A divided bike lane with speed bumps (Maus, 2022)</i>	● Relatively inexpensive. ● Easy installation.	stability. ● Can reduce travel time of cyclists/skate boarders due to reduced speeds. ● Can cause cyclist collisions due to reduced stability.	Grooves do not cause as many stability issues as speed bumps do.
4	 <i>A green bike lane (Adriazola, 2022)</i> Painting Bike Lanes Green: By painting the lanes green and clearly marking where bikers can and cannot go, it will clear up traffic congestion.	● Affordable method that can be used everywhere ● Aesthetically pleasing ● Works without any power draw	● Needs to be repainted every few years ● Doesn't physically prevent accidents from people violating ground markers	Add painted lane lines while including physical barriers as well, this will allow both bikers to use it, allow it to be cheap, and allow it to keep them separate so there is no interaction between the two that can cause danger.

c. Sketching

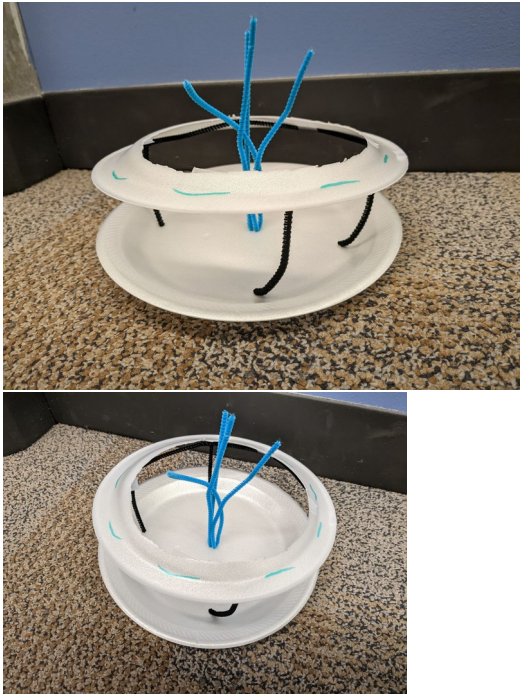
Table 6. Generating ideas with sketching

#	Sketch	Solution name	Unique aspects, brief description
1	Proposed new bike lane that encompasses the engineering fountain	Engineering fountain bike lanes	Unique circular encompassing design. Goes around the fountain completely and

	 A drawing of the bike lane (Self, 2022)		allows bikers to continue onward at increased speed rather than having to slow down at left and right intersections.
2			
3			
4			

d. Low Fidelity Prototyping

Table 7. Generating ideas with low fidelity prototyping

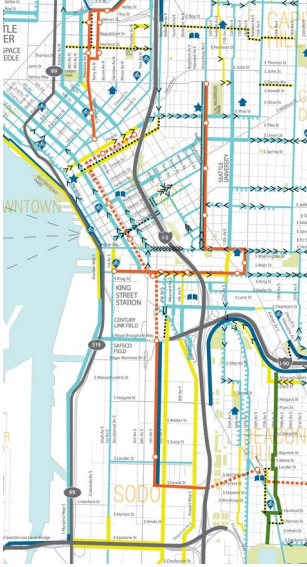
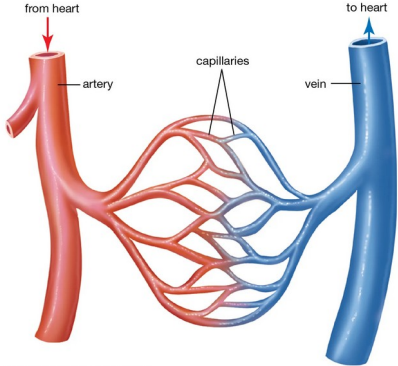
#	Prototype picture	Solution name	Unique aspects, brief description
1	Prototype of raised bike lane above fountain  A model of the bike lane (Self, 2022)	Raised bike & pedestrian lane	Unique design that features a raised area for bikers or walkers. Quite an influential 3d look at a method that reduces traffic. This model focuses on bringing out the subduction idea of a raised platform for bikers or walkers.

2			
3			
4			

e. Biomimicry

Table 8. Generating ideas with biomimicry

#	Sketch/picture/image	Solution name	Unique aspects, brief description
1	 <i>Underground Bike tunnel (Andersen, 2022)</i>  <i>Ant hill design inspiration (AntKeepers, 2021)</i>	Ant-hill like Interconnected bike lane tunnel.	Using an ant hill as inspiration, Purdue could build more tunnels for pedestrians, allowing for bikers to more freely use the bike lanes.

2	 <i>Blood Vessel like City Bike Lanes</i> (Fucoloro, 2014)  © 2013 Encyclopædia Britannica, Inc. <i>Blood vessel</i> (The Editors of Encyclopaedia Britannica, 2022)	Bike lane blood vessel exclusivity.	Taking inspiration from blood vessels, the bike lanes would be exclusive throughout the intersection, only allowing for bikes to cross. This is like blood vessels since they keep everything on the inside and outside separated.
3	 <i>Reflective beetles</i> (Libby, 2011)	beetle shell like Reflective barriers	Bioluminescence is a trait found throughout nature in many animals which helps with their visibility to warn predators.

f. Thought Experiments with Pros and Cons Evaluation

Table 9. Thought experiments with pros and cons evaluation

#	Design Alternative (include a name and an image)	Pros (advantages)	Cons (disadvantages)
1	Yield Signs  <i>Pedestrian Sign (Pedestrian Plan Ideas, 2022)</i>	● Inexpensive ● lasts for long periods of time ● reflective sign can be seen at night	● Does not physically prevent pedestrians and bikers from interacting. ● Forces bikers to slow down for pedestrians which slows down intersection traffic.
2	Adding a proper stoplight  <i>A stoplight in an intersection (Santos, 2022)</i>	● Would help traffic flow ● would completely prevent any accidents from occurring ● no more injuries could occur	● would slow down traffic ● is extremely expensive ● would decrease the look of campus
	adding a stop sign to the intersection  <i>A stop sign in a neighborhood (Aronczyk & Horowitz-Ghazi, 2021)</i>	● also inexpensive ● will help create order for the intersection ● will prevent accidents	● the sign will most likely not be followed ● does not make as much sense as a yield sign in the situation ● will seem out of place in the center of campus

V. ITERATION #1

Table 10. Responding to feedback

#	Based on feedback/data from ... about	We made changes in...by...
1	Based on feedback data from an anonymous Purdue freshman student about our new 360 degree painted bike lane around the fountain we made sure that instead of just using paint, we incorporated a bike lane with a separated barrier that would improve both visibility and separation between bikers and walkers.	We made changes to the design by incorporating slightly raised concrete barriers around the outside of the fountain ring. We also will make sure to include clear indicators of where the bike lane starts and stops so that it is easily accessible to any potential user.
2	Based on feedback from another group about our fountain intersection improvement project, we made sure to improve our project by doing everything possible to mitigate the extreme speeds of bikers . The feedback stated that they are taking turns at extremely fast speeds and are quite dangerous, and that it would be very bad for someone to get hit.	We made changes to our design by adding in rumble strips into the bike lane. This will allow the bikers to move at speeds that are more accessible to everyone, and will prevent them from going way too fast and at dangerous speeds. This improvement will not only make it clear where the bike lane is, but it will also make the intersections safer.

- Illustrate
 - We were able to use feedback from both our experimenting with different models and the feedback we received from the student and the other team in our process by adding new features, such as the rumble strip, the physical barrier, the painted lanes, and other innovations that we hadn't thought of before in the process. Through this added feedback, we were able to create a design that is much better than our original thought of the idea.
- Reflect
 - Our understanding of the problem has greatly increased from the feedback we have gained. At first, we knew that we would frequently get into close calls with bikes and other vehicles, but we did not know how widespread the issue was. In our team, we have not changed our management or teaming process, but we have made changes to our design process for sure. We made sure that we consult our clients and any other users of our product before making any large decisions. We also make sure to think about how we can improve the product when we are not able to ask stakeholders at the time. This allows us to continue our design process timely, whilst also staying grounded in the reality of what we are creating.

VI. PROTOTYPING, TESTING AND WDM

a. Prototypes

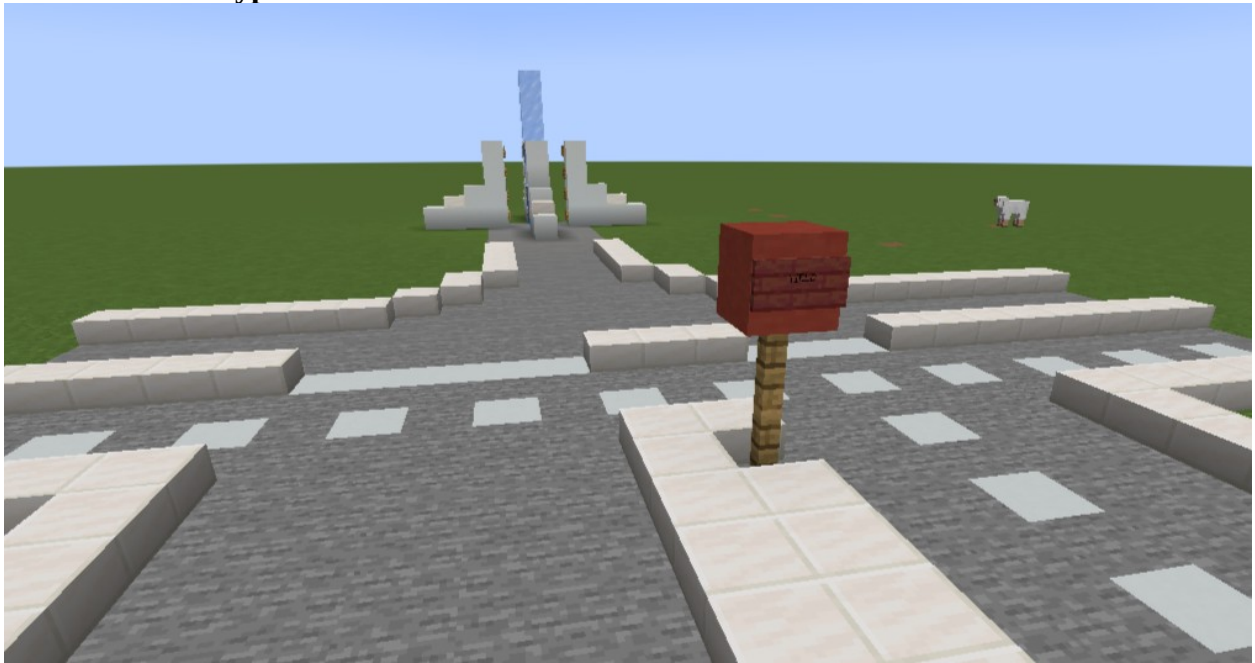


Figure 3. Yield Sign for Bikers Prototype in Minecraft with Permanent Signage



Figure 4. Bike lane Prototype in Minecraft with Permanent Tall Plastic Barriers

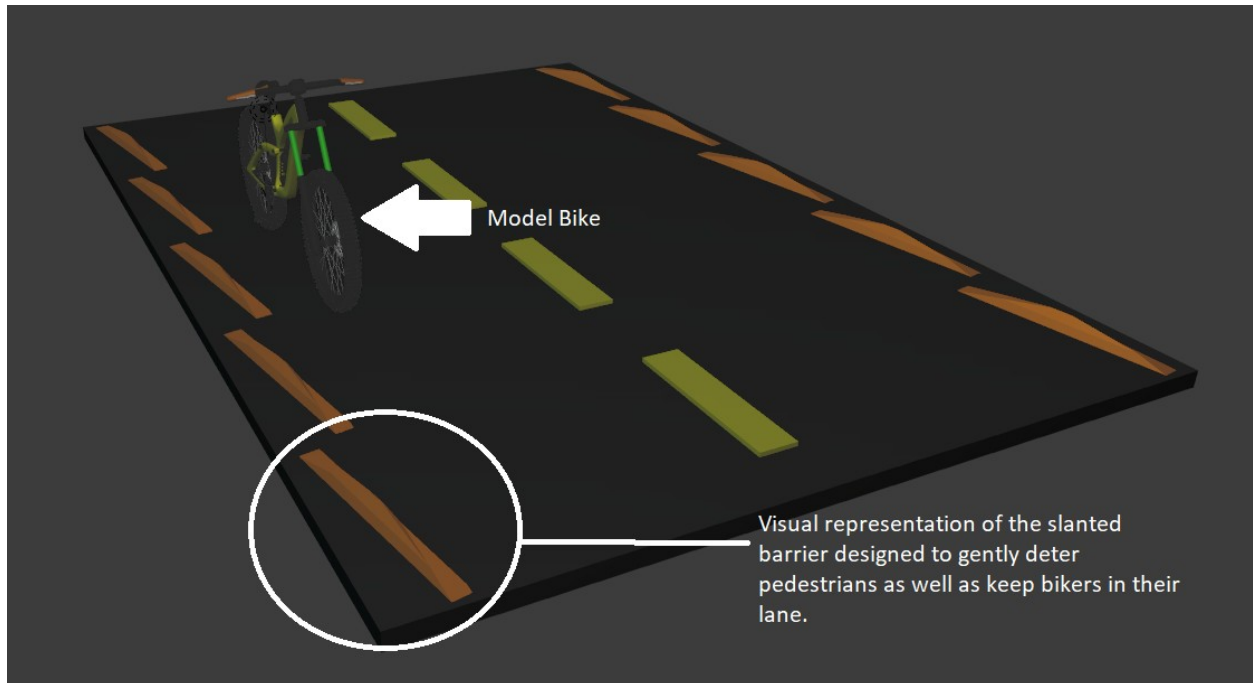


Figure 5. Bike lane model with protective barriers on the sides to deter pedestrians

b. Prototype Testing Protocol

Prototype 1:

The first prototype features a large red sign that says “yield.” This yield sign will force bikers to slow down and stop whenever pedestrians are in the way. This will slow down traffic, but will prevent additional accidents from occurring. We tested this in minecraft by running a second traffic simulation. We put villagers in carts and had others walk in the walking lane. using different combinations of both, we tested to see which were faster and which were slower. We found that the safety of the yield sign is unparalleled. In the testing, there were zero close calls between bikes and pedestrians when the yield sign was in place. there were many more close calls when the sign was removed. This sign proved quite important for keeping the intersection safe. Although it increased the safety, it greatly decreased the speed of the intersection. This means that it is not nearly as efficient as prototype two or three. This is not a good thing at all, as we want to keep the intersection at its speed, whilst also increasing safety, rather than slowing down the intersection and causing traffic jams.

Prototype 2:

Our testing protocol included running different types of pedestrians in minecraft through our bike lane and seeing the different interactions they had. We used three main situations. We had one “biker” interact with one pedestrian, two “bikers” interacting with each other by going in different directions, and finally a mix of both bikers and pedestrians walking together. running these simulations, with and without the bike lanes yielded different results. We found that there were many fewer collisions, and it was much safer when the new barriers were included. We measured this noticed safety by counting the number of collisions, which decreased with the new

barriers. Another design requirement, increasing the flow of traffic, was also considered in the testing. We found that no matter the intersection, having dedicated lanes increased the speed in which everyone could travel. This increase was dramatic, as without proper marking, constant sidesteps and waiting for a bike to pass slowed down the movement of traffic.

Prototype 3:

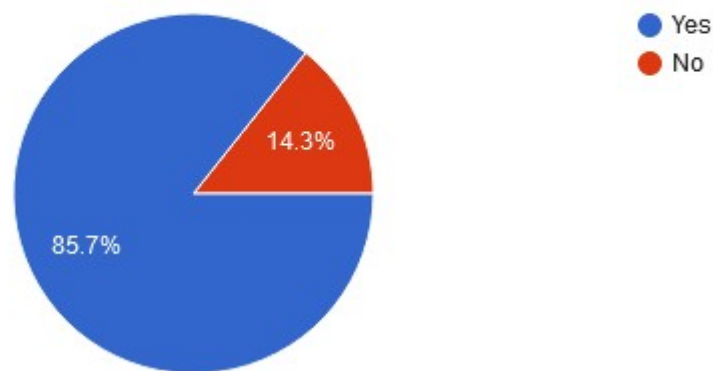
Our third prototype simulated a potential solution to keep pedestrians out of the bike lane while also making sure the bikers stay within the bike lane. This solution is ideal since it is low cost and simple to install. This solution allows for the addition of low, slanted barriers along the edge of the bike lanes, as shown in the 3-D model. Through the addition of these barriers, pedestrians were able to successfully stay out of the bike lane at a higher rate, while bikers would tend to remain in the bike lanes. The amount of collisions between bikers and pedestrians significantly decreased with the addition of this low cost barrier. The multiple test trials definitively proves that through the addition of these barriers, the amount of collisions significantly decreased. In the simulations tested, this solution did not slow down the movement of traffic, however it did not increase the flow of traffic either; the traffic rate remained stagnant.

c. Test Results for Top 3 Alternatives

Survey Outcome for Enclosed Bike Lane Prototype:

Is this Safe?

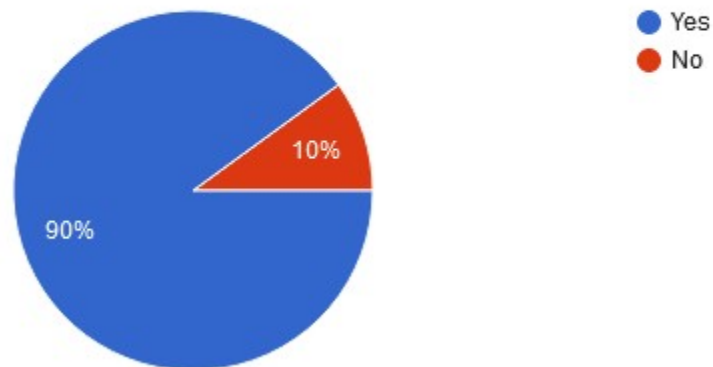
21 responses



Survey Outcome for Yield Sign Prototype:

Is this safe?

20 responses



Survey Outcome for Small Speed Bump Barrier Prototype:

Is This Safe?

21 responses

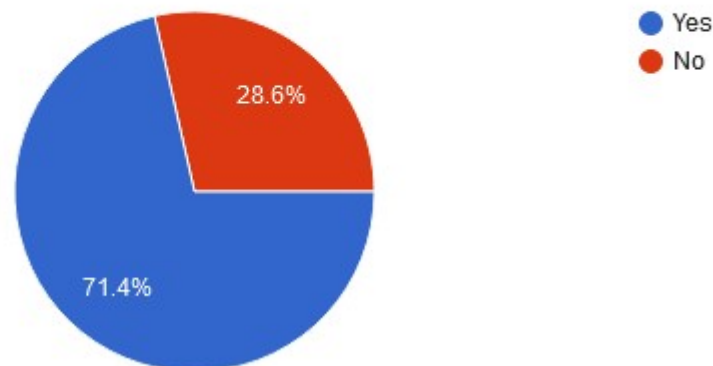


Figure 6. Testing for Safety

Prototype 1:

Prototype one features a yield sign. This added signage increased the perceived safety of the intersection and was overall taken well by the anonymous surveyors, with 85.7% believing that the intersection is safe with the added yield sign.

Prototype 2:

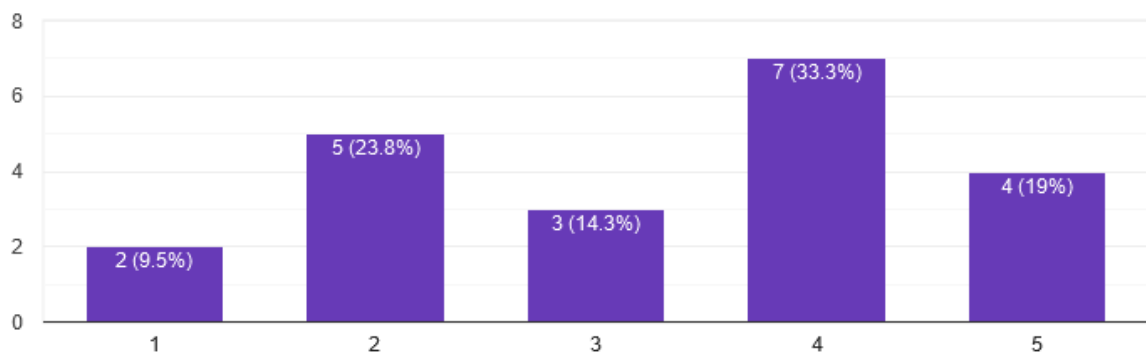
Prototype two features a divided bike lane. These added barriers will greatly increase the safety of the intersection, and the results agree with this assertion. with 90% of people believing that this design was safe. This redesign is by far the most advanced and safe option, as it is quite difficult to run into a pedestrian when enclosed in your own lane.

Prototype 3:

Prototype three was much more controversial. The bumps in the road were much more hazardous for both pedestrians and bikers, but would be much cheaper than the full sized barrier alternatives. This led to a split in the feedback, with only 71.4% of people thinking that this attempt was safe. A trade off of almost 20% for a decrease in price is not nearly worth it. Although the idea was promising, according to the people, it is not nearly safe enough to be used at Purdue.

Survey Outcome for Enclosed Bike Lane Prototype:

21 responses

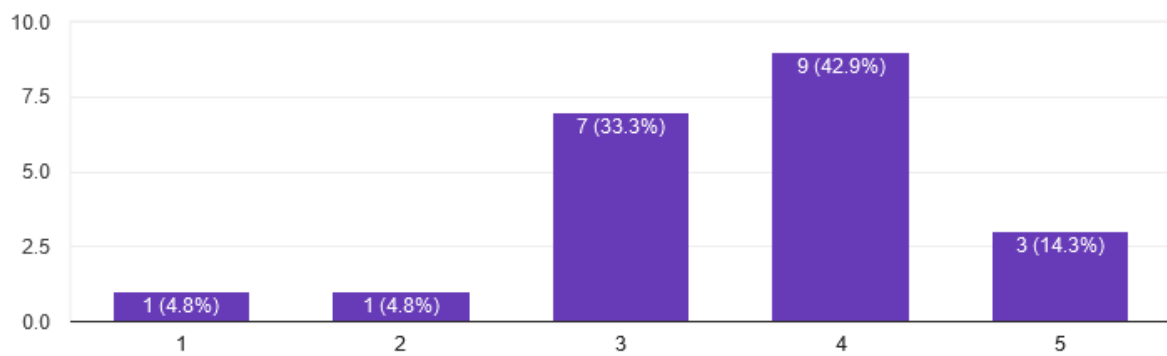


Survey Outcome for Yield Sign Prototype:

Is This Convenient?

 Copy

21 responses



Survey Outcome for Small Speed Bump Barrier Prototype:

Is This Convenient?



21 responses

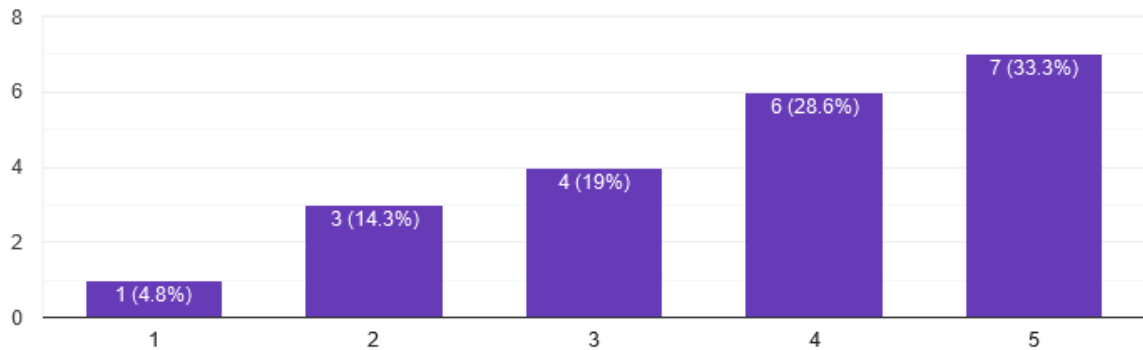


Figure 7. Testing for Convenience

Prototype 1:

Prototype one was quite indecisive. when looking at the spread, there was no set decision. This makes sense, as to some (pedestrians) having all of the bikes stop for you is extremely convenient. As a biker, it is much less convenient. Having to stop for every single pedestrian is extremely inefficient and inconvenient.

Prototype 2:

Prototype two takes a different approach. This figure stood out, as it was centered much more toward 3 and 4. This is because this approach is much more manageable for both bikes and walkers when sharing an intersection. Having both have a dedicated area allows them to live at peace with mild discomfort, which explains the slight trend from 5 toward lower numbers.

Prototype 3:

Prototype 3 is quite an enigma. There is a distinct trend of people choosing that it is of highest convenience, whilst also calling it the most unsafe option. This is quite puzzling, as there should be at least a slight correlation between the two, but the opposite is true. It seems that the idea of having a low invasive barrier is quite convenient to all, and that is why it received the scores that it did. I am quite unsure why there was such a range, but I believe that has to do with the fact that the installation would be quite quick and non-invasive as well.

d. Weighted Decision Matrix (WDM)

Weight/Importance: 15: Must have 10: Good to have 5: Nice to have		BENCHMARKING 5: Best, 1: Worst						
		BENCHMARKING (Unweighted)			BENCHMARKING (Weighted)			
DESIGN REQUIREMENTS (Criteria & Constraints)		METRICS						
		Weight/Importance	Yearly Cost for Maintenance	Setup/Installation Cost	Monthly Number of Accidents	Additional Waiting Time for Pedestrians and Bikers	Feedback from Pedestrians	Feedback from Bikers
1 Affordability		10	X	X				
2 Ease of Maintenance		10	X					X
3 Safety		15			X		X	X
4 Convenience		5				X	X	
ENGINEERING TARGETS -->			<100	<3000	<15	<2		<2
		Units	\$	\$	Number	Minutes	-	-

		important to the Purdue students, as the engineering fountain is something that they are quite proud of, and disturbing the area with bright colored barriers is not a good idea.
2	Based on feedback from team fifteen about the distance between the individual barriers to enhance safety,	We made changes by increasing the distance between each barrier. Increasing this distance will allow bikers to swerve out of the way safely if an accident was about to occur. This increase in distance would also decrease the cost of our solution, which is something important to the university, because installing a safe set of barriers affordably is much preferred to expensive barriers.
3	Based on feedback from team one about the material used in the barriers, we had to reevaluate the specific plastic composite to reduce potential injuries.	Based on this feedback, we decided to research specific types of plastic composites which could be used for the creation of the barriers. We settled on high density polyethylene (HDPE), since it is a weather resistant plastic which is not dangerous to run into. Although bikers might still get injured from use of this material, this material is used in areas like playgrounds, where a safe plastic is needed.
4	Based on feedback from team fifteen about the high cost to implement the raised barriers,	Our decision to use high density polyethylene to create our barriers will both be affordable and produce high quality and well finished barriers that are able to withstand all weather conditions in West Lafayette.

- Reflecting on the Problem

Figuring out an effective solution for increasing the safety of the Engineering Fountain is quite difficult. Through inter and outer-team discussion, we have come to a common consensus. Over time, with our surveys and research, we have realized that barriers are the ideal. This is because the yield sign will not be followed by certain bikers and pedestrians. We also realized that speed bumps in the ground are not ideal as they would be hidden by the snow. These multiple cons for each design made it quite clear that the raised barriers are ideal.

VIII. OVERVIEW OF FINAL DESIGN

a. Detailed Design

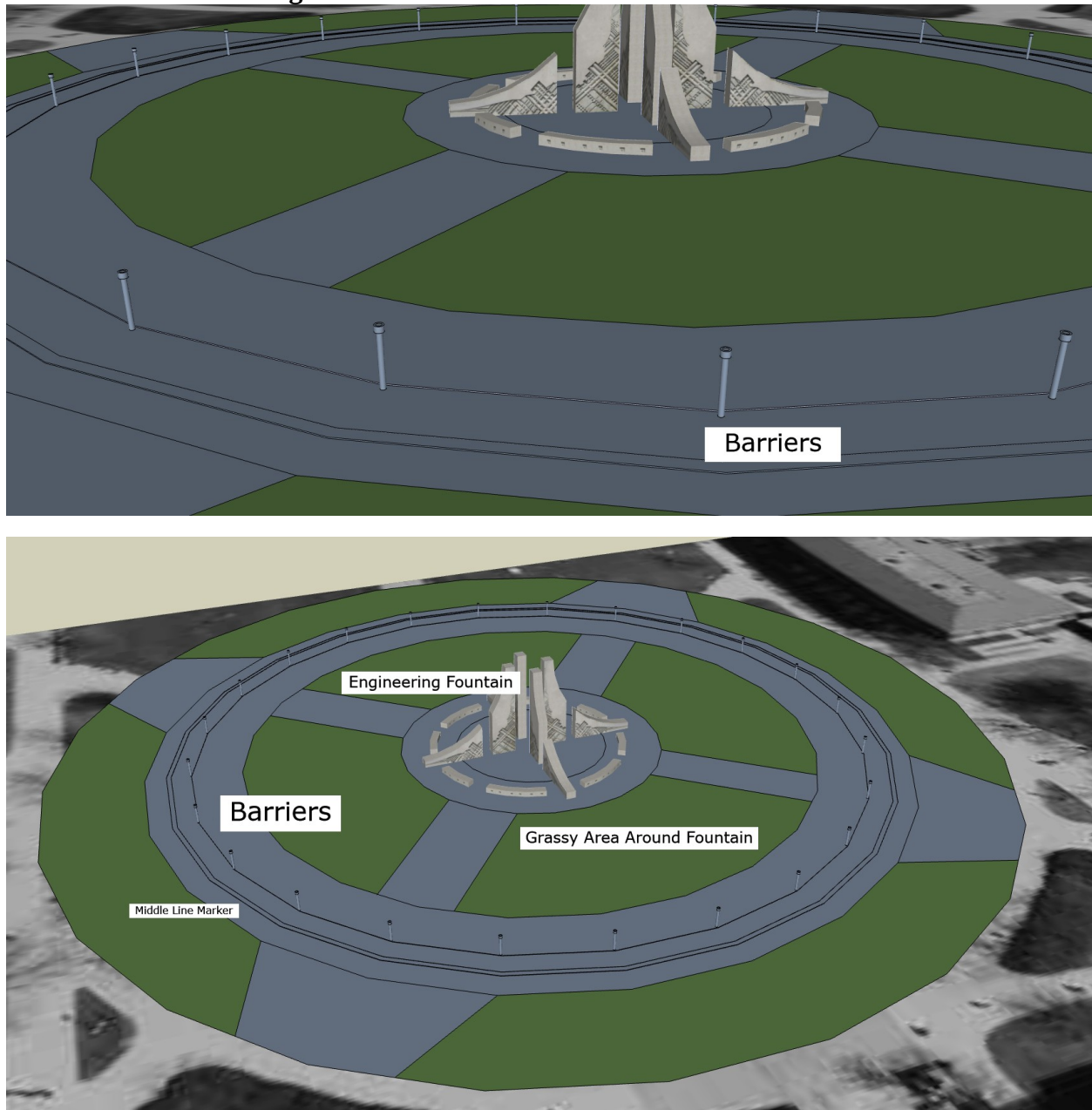


Figure 9. Computer-aided prototype of Engineering Fountain Intersections

This CAD design features many of the proposed barriers that could be added to the intersections to improve safety. Featured in the model is an accurate representation of the fountain surrounded by the grassy areas and most importantly the barriers. The color of the barriers is a Purdue gray color, with the top tipped with the signature Purdue gold. This could not be represented in the 3d model software, as it does not include the ability to color models in that way. There are multiple lines that also separate the two directions at which someone can travel through the bike lane.

b. Data on Final Solution

The feedback from bikers and pedestrians about the safety and convenience of our final solution is shown in Figures 10 and 11. Over 85% of respondents feel that this solution is safe and over 52% of respondents gave a rating of 4 or higher for convenience (a score of 1 refers to “not convenient” while 5 refers to “very convenient”). Further, the barriers mostly prevent physical interaction between pedestrians and bikers, therefore, reducing the number of accidents by a great extent. Additionally, in the winter months, snow is present on and around the bike lanes. Since the barriers which would be used are quite tall and bright, they would still be completely visible to bikers and pedestrians, therefore, preventing either one of them from crashing into the barriers themselves. These aspects contribute greatly towards increasing the safety of the final solution, as indicated by the highest weighted score of 60 for safety in the weighted decision matrix (Figure 12). Moreover, the weighted decision matrix shown in Figure 12 also highlights that overall, installing barriers along the bike lanes is the best option as it has the highest weighted score of 116.7 points.

Although this design is not the least expensive option, it is very safe, convenient, weatherproof, and requires infrequent maintenance proving that it meets the design requirements and is the best overall solution.

Is this Safe?

21 responses

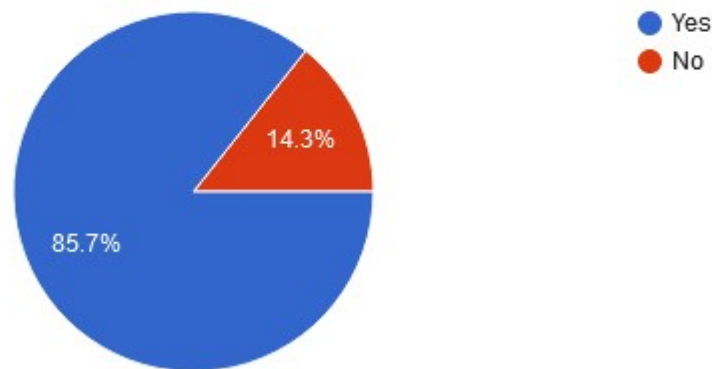


Figure 10. Public Appeal Testing for Safety

21 responses

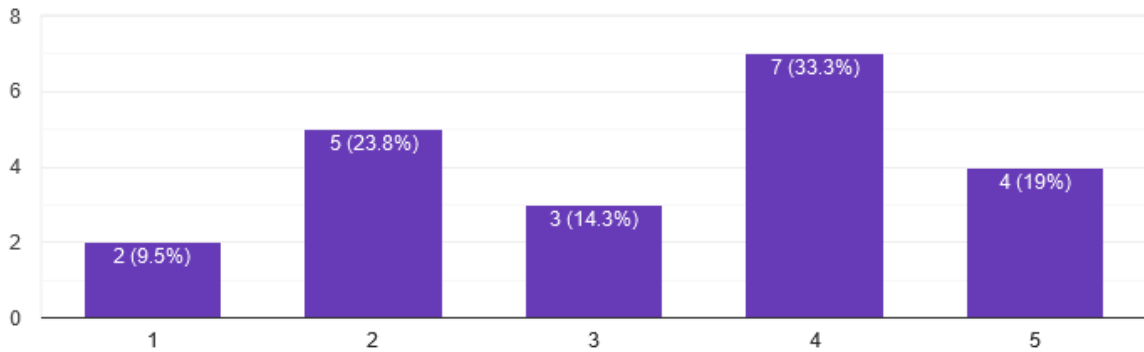


Figure 11. Public Appeal Testing for Convenience

BENCHMARKING (Weighted)			
DESIGN REQUIREMENTS (Criteria & Constraints)			
	Yield sign	Plastic barriers	Rumble strips
1 Affordability	10	10	22.5
2 Ease of Maintenance	30	25	30
3 Safety	50	60	40
4 Convenience	16.67	21.67	16.67
ENGINEERING TARGETS -->			
	106.7	116.7	109.2

Figure 12. A section of the weighted decision matrix

c. Design Optimization

Throughout the engineering process for our project, we improved our final design through combining multiple ideas so that our final design was the most optimal solution. By using high density polyethylene (HDPE), a cost effective material, the price of our solution was able to be drastically reduced. Furthermore, the barriers were efficiently spaced in a specific format, so that they were far enough to reduce the cost further, but also close enough to deter pedestrians from the bike lane. This optimization was able to be implemented through a combination of peer feedback, research, and our previous prototype ideas. Through trial and error, our final design was able to contain the most optimal cost and efficacy.

d. Novel Aspects of the Solution

Our design implementation consists of the installation of barriers in a strategic manner that allows for the intersection by the engineering fountain to be safer for both students and staff. Barriers are implemented throughout areas and intersections all over the world. What makes our barriers different from some of these other barriers is the way that they are placed. First of all, we wanted our barriers to be specific to Purdue University. So our barriers will be colored in Purdue

University colors, in order to represent the student body and pride. Additionally, and as we have stated before, our barriers will be placed so they are not an interruption to the flow of traffic, just placed so that they divide bikers and skateboarders from pedestrians. A lot of barriers around the world are placed to disrupt and stop the flow of traffic, but ours are not implemented for that reason.

e. Trade-off Decisions and Limitations of the Solution

Our design does not have any form of active discouragement from disobeying the rules, therefore it is still possible for bikers to ride on the pedestrian lane and vice-versa. Therefore, there is still a chance of accidents happening and people getting injured, although a lot less than if the barriers were not installed. To minimize the impact of this limitation, Purdue students can be familiarized with the traffic rules and encouraged to follow them.

Moreover, adding barriers to the bike lanes would mean that bikers who have to navigate through these barriers have to swerve through them while turning and giving way to pedestrians crossing the bike lane. This could be dangerous as some bikers could find it difficult to do so. To counter this problem, Purdue could offer bike lessons that teach how to ride the bike in certain difficult situations so that accidents and injuries can be avoided.

f. Lessons Learned

Throughout our engineering project process, there were many strengths and weaknesses to our methodology. One large strength that our group had was our ability to get work done. Together, using the time in class, we were able to complete our assignments and use our time wisely. Using this process, we were set up quite well to succeed on this project. Another strength about our process was our ability to follow the rubric and create good work. Sometimes, when working on difficult assignments, work can be made of low quality. Throughout our assignment work, there was never a time where we created low effort work, we always gave it our all. A weakness that we had is that we had trouble with the ability to meet outside of class. Our schedules typically conflict with each other, so it is quite important that we got work done in class because it was so hard to meet outside of class. One other problem we had was deciding an intersection. At first we were quite divided on which intersection to choose. Over time, we found a great project to work with, but at first it was quite difficult to decide.

By working with the group, we all grew in our team building skills. This skill is something that we will all use throughout our careers, so it is extremely crucial that we learn it early in our engineering class. We also all learned communication skills. Constantly keeping in touch and learning different methods in which we can contact each other for maximum efficiency is quite important, and is something that is beneficial to all of us.

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X. APPENDICES